MILITARY SPECIFICATION SHEET

SWITCHES, CODE INDICATING WHEEL (PRINTED CIRCUIT), THUMBWHEEL SWITCH ASSEMBLY, INTEGRAL BEZEL, NONSEALED, NONILLUMINATED, SOLID MOUNTING, STYLE SRPC29

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22710.

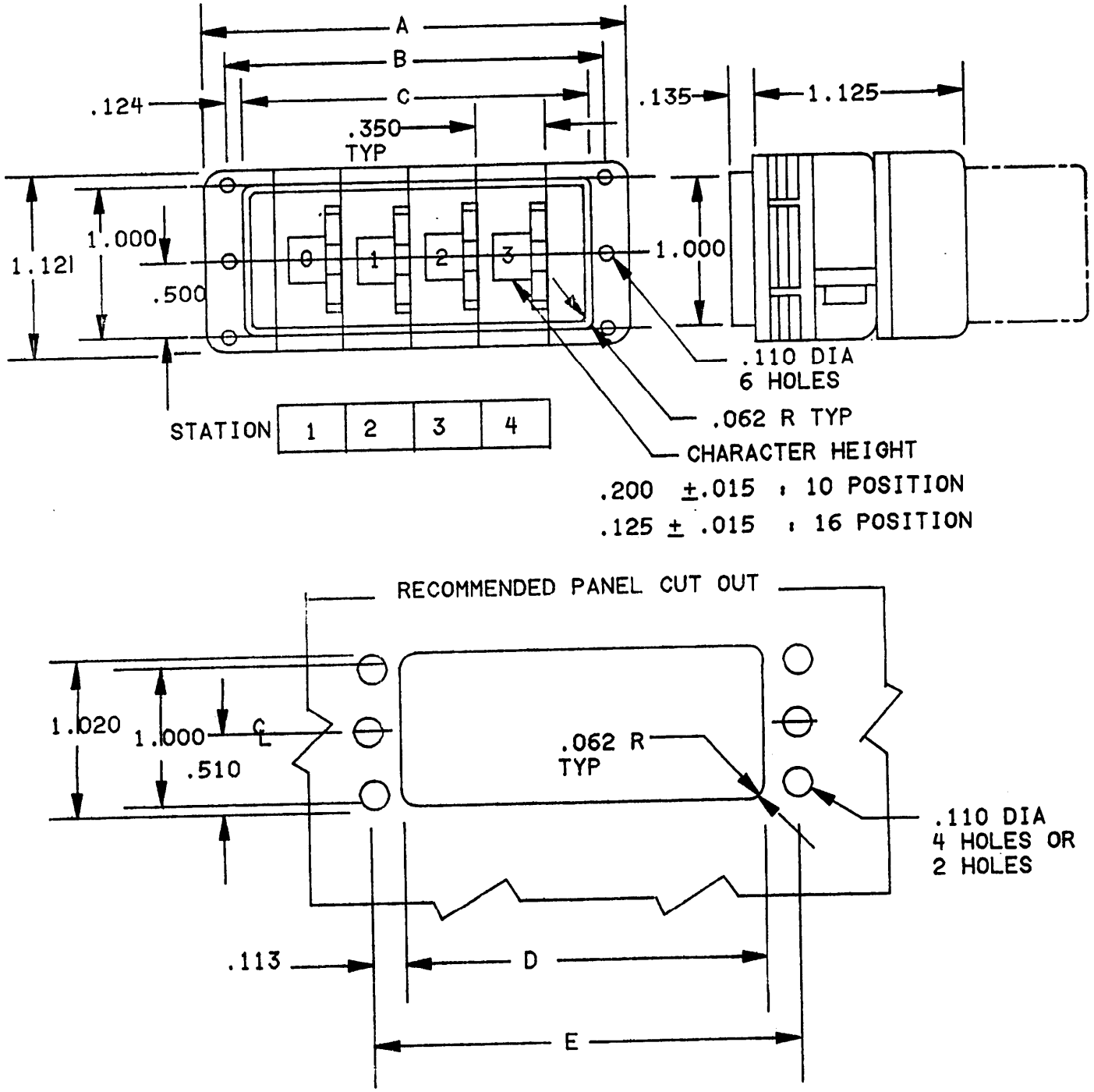


FIGURE 1. Style SRPC29 switch and panel cutout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

FSC 5930

Number of Modules	Dimensions with RFI Shield				
	A	B	C	D	E
1	1.11 (28.19)	.914 (23.22)	.668 (16.97)	.688 (17.48)	.914 (23.22)
2	1.46 (37.08)	1.266 (32.16)	1.020 (25.91)	1.040 (26.42)	1.266 (32.16)
3	1.81 (45.97)	1.618 (41.10)	1.372 (34.85)	1.392 (35.36)	1.618 (41.10)
4	2.16 (54.86)	1.970 (50.04)	1.724 (43.79)	1.744 (44.30)	1.970 (50.04)
5	2.51 (63.75)	2.322 (58.98)	2.076 (52.73)	2.094 (53.19)	2.322 (58.98)
6	2.87 (72.90)	2.674 (67.92)	2.428 (61.67)	2.448 (62.18)	2.674 (67.92)
7	3.22 (81.79)	3.026 (76.86)	2.780 (70.61)	2.800 (71.12)	3.026 (76.86)
8	3.57 (90.68)	3.378 (85.80)	3.132 (79.55)	3.152 (80.06)	3.378 (85.80)
9	3.92 (99.57)	3.730 (94.74)	3.484 (88.49)	3.504 (89.00)	3.730 (94.74)
10	4.27 (108.46)	4.082 (103.68)	3.836 (97.43)	3.856 (97.94)	4.082 (103.68)

FIGURE 1. Style SRPC 29 switch and panel cutout - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

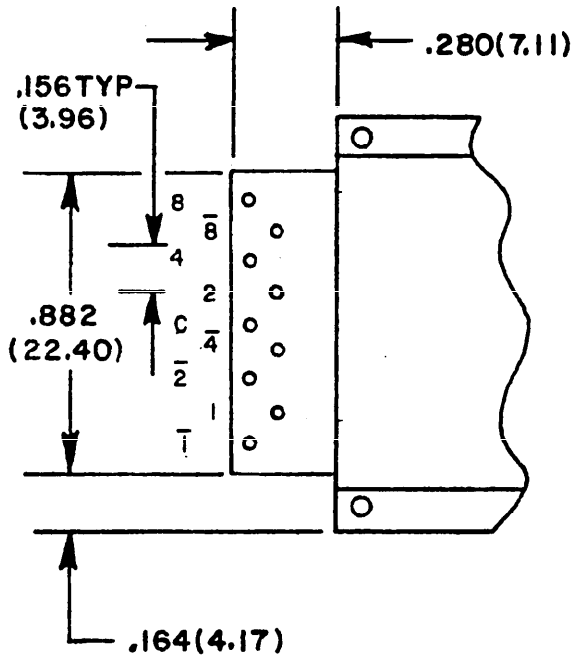
INCHES	MM
.015	.38
.062	1.57
.110	2.79
.113	2.87
.121	3.07
.124	3.15
.125	3.18
.135	3.43
.200	5.08
.350	8.89
.500	12.7
.510	12.9
1.000	25.4
1.020	25.9
1.21	30.7

Number of Modules	Dimensions without RFI shield				
	A	B	C	D ±.006 (.15)	E
1	1.10 (27.9)	.910 (23.1)	.662 (16.8)	.684 (17.4)	.910 (23.1)
2	1.45 (36.8)	1.260 (32.0)	1.012 (25.7)	1.034 (26.3)	1.260 (32.0)
3	1.80 (45.7)	1.610 (40.9)	1.362 (34.6)	1.384 (35.2)	1.610 (40.9)
4	2.15 (54.6)	1.960 (49.8)	1.712 (43.5)	1.734 (44.0)	1.960 (49.8)
5	2.50 (63.5)	2.310 (58.7)	2.062 (52.4)	2.084 (52.9)	2.310 (58.7)
6	2.85 (72.4)	2.660 (67.6)	2.412 (61.3)	2.434 (61.8)	2.660 (67.6)
7	3.20 (81.3)	3.010 (76.5)	2.762 (70.2)	2.784 (70.7)	3.010 (76.5)
8	3.55 (90.2)	3.360 (85.3)	3.112 (79.0)	3.134 (79.6)	3.360 (85.3)
9	3.90 (99.1)	3.710 (94.2)	3.462 (87.9)	3.484 (88.5)	3.710 (94.2)
10	4.25 (108)	4.060 (103.1)	3.812 (96.8)	3.834 (97.4)	4.060 (103.1)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses or in table.
3. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
4. Unless otherwise specified, tolerance is ±0.010 (0.25) for three place decimals.
   ±0.030 (0.75) for two place decimals.

FIGURE 1. Style SRPC29 switch and panel cutout - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.



BOARD PLATED ON BOTH SIDES
WITH PLATED THRU HOLES FOR
SOLDERED WIRE TERMINATION

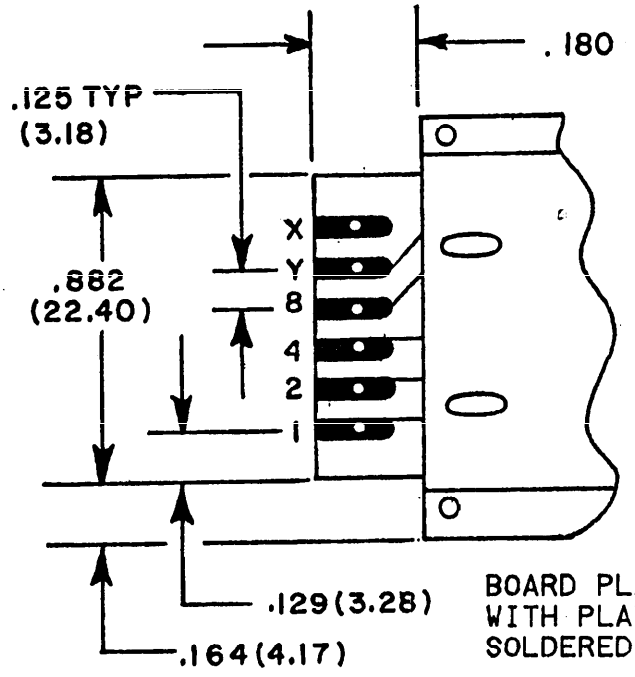
HOLES: 0.040(1.02) DIA

MODULE A AND H

TRUTH TABLE (CIRCUIT CONFIGURATION 1) 36° INDEXING								
DIAL MARKING	COMMON C CONN. TO							
	1	2	4	8	$\overline{1}$	$\overline{2}$	$\overline{4}$	$\overline{8}$
0					●	●	●	●
1	●					●	●	●
2		●			●		●	●
3	●	●					●	●
4			●		●	●		●
5	●		●			●		●
6		●	●		●			●
7	●	●	●					●
8				●	●	●	●	
9	●			●		●	●	

FIGURE 2. Module and circuit configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.



BOARD PLATED ON BOTH SIDES WITH PLATED THRU HOLES FOR SOLDERED WIRE TERMINATION

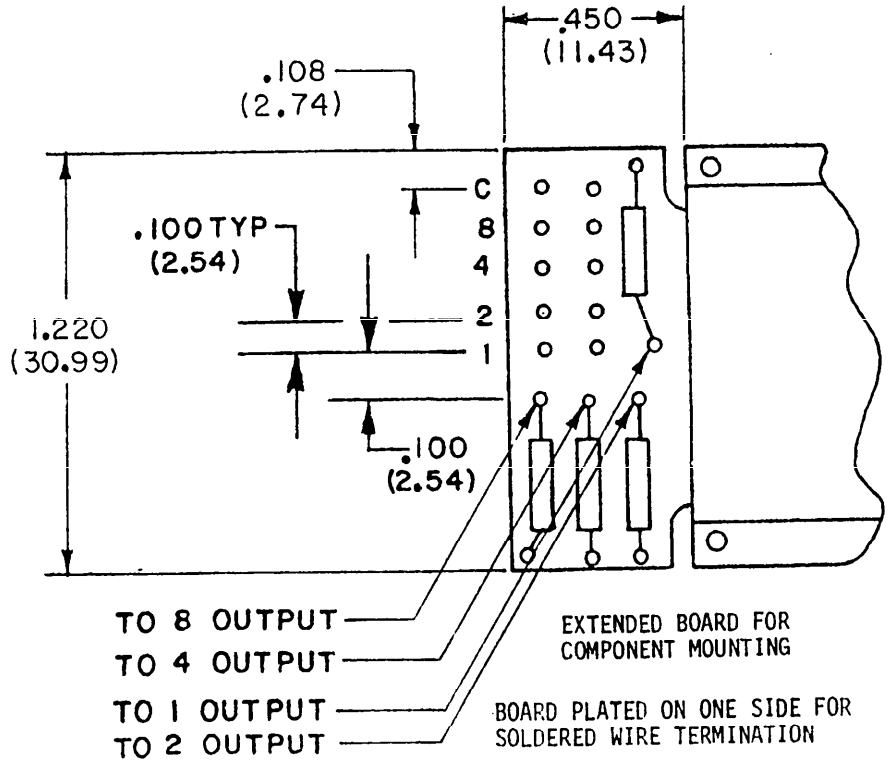
HOLES: 0.040 (1.02) DIA

MODULE B AND J

TRUTH TABLE (CIRCUIT CONFIGURATION 2) 36° INDEXING					
DIAL MARKING	CONN.X (●)& Y (O) CONN. TO				
		1	2	4	8
0		O	O	O	O
1		●	O	O	O
2		O	●	O	O
3		●	●	O	O
4		O	O	●	O
5		●	O	●	O
6		O	●	●	O
7		●	●	●	O
8		O	O	O	●
9		●	O	O	●

FIGURE 2. Module and circuit configuration - Continued.

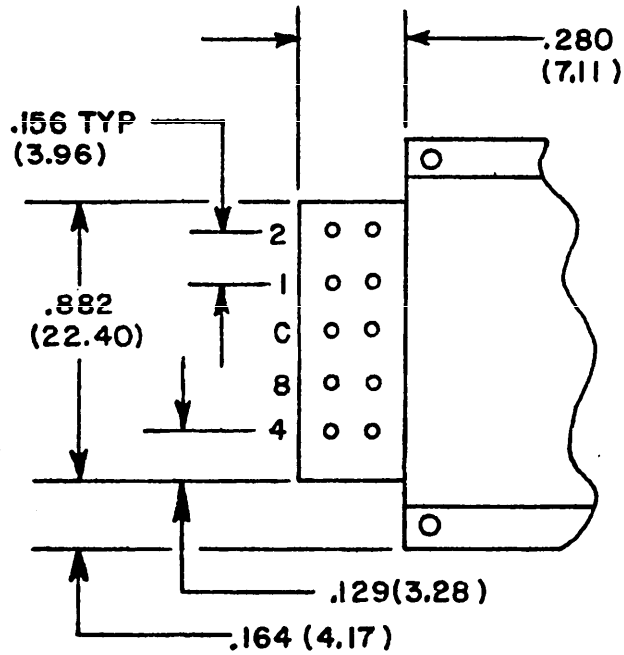
Source: https://assist.dla.mil -- Downloaded: 202610-19T13:35Z
Check the source to verify that this is the current version before use.



MODULE C AND K

HOLES, COMPONENT - 0.046 (1.17) DIA

WIRING - 0.046 (1.17) DIA



BOARD PLATED ON BOTH SIDES
WITH PLATED THRU HOLES FOR
SOLDERED WIRE TERMINATION
HOLES: 0.049(1.24)DIA

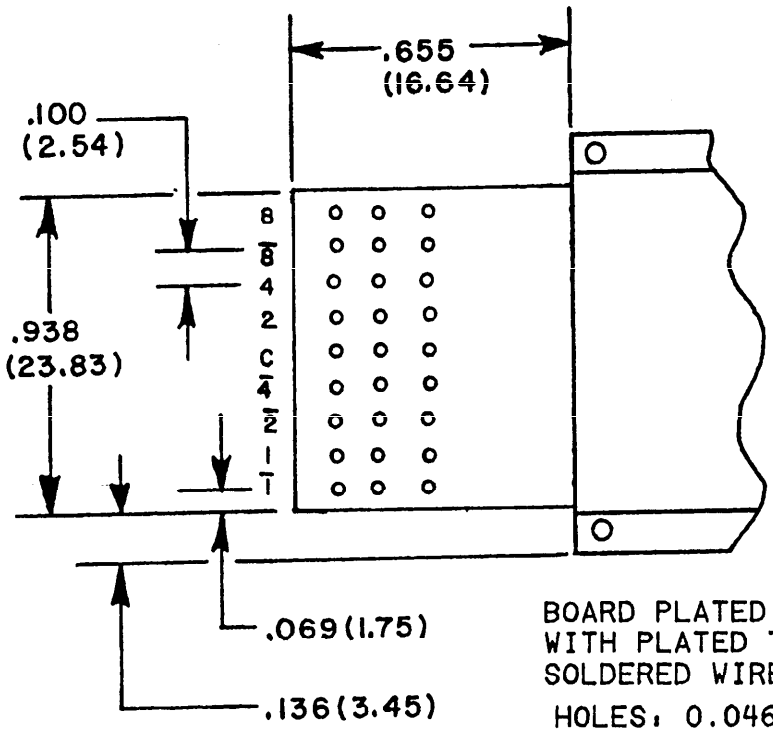
MODULE D AND L

TRUTH TABLE
(CIRCUIT CONFIGURATION 3)
36° INDEXING

DIAL MARKING	COMMON C CONN. TO				
		1	2	4	8
0					
1		●			
2			●		
3		●	●		
4				●	
5		●		●	
6			●	●	
7		●	●	●	
8					●
9		●			●

FIGURE 2. Module and circuit configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.



BOARD PLATED ON BOTH SIDES
WITH PLATED THRU HOLES FOR
SOLDERED WIRE TERMINATION
HOLES: 0.046(1.17) DIA

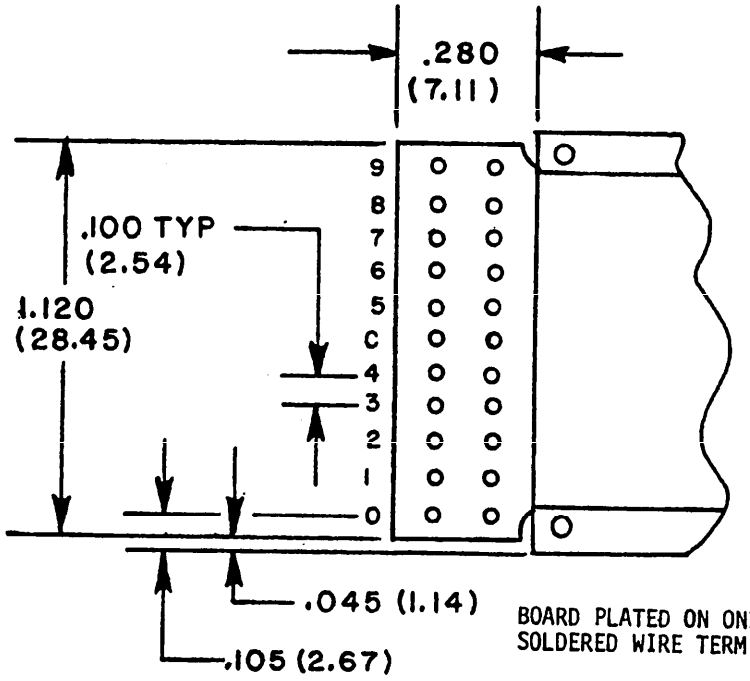
MODULE E AND M

TRUTH TABLE
(CIRCUIT CONFIGURATION 4)
36° INDEXING

DIAL MARKING	COMMON C CONN. TO							
	1	2	4	8	$\overline{1}$	$\overline{2}$	$\overline{4}$	$\overline{8}$
0					●	●	●	●
1	●					●	●	●
2		●			●		●	●
3	●	●					●	●
4			●		●	●		●
5	●		●			●		●
6		●	●		●			●
7	●	●	●					●
8				●	●	●	●	
9	●			●		●	●	

FIGURE 2. Module and circuit configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

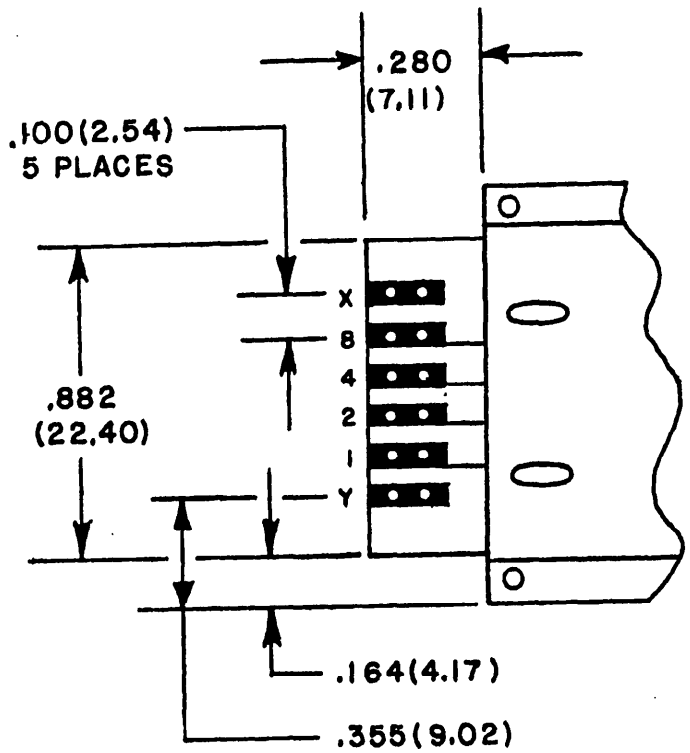


MODULE F AND N

TRUTH TABLE (CIRCUIT CONFIGURATION 5) 36° INDEXING	
DIAL MARKING	COMMON C CONN. TO
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

FIGURE 2. Module and circuit configuration - Continued.

Source: https://assist.dla.mil - Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.



BOARD PLATED ON BOTH SIDES WITH PLATED THRU HOLES FOR SOLDERED WIRE TERMINATION

HOLES: 0.046(1.17) DIA

MODULE G AND P

TRUTH TABLE
(CIRCUIT CONFIGURATION 6)
22.5° INDEXING

DIAL MARKING	COMMON X (●) COMMON Y(O) CONN. TO				
		1	2	4	8
0		O	O	O	O
1		●	O	O	O
2		O	●	O	O
3		●	●	O	O
4		O	O	●	O
5		●	O	●	O
6		O	●	●	O
7		●	●	●	O
8		O	O	O	●
9		●	O	O	●
10		O	●	O	●
11		●	●	O	●
12		O	O	●	●
13		●	O	●	●
14		O	●	●	●
15		●	●	●	●

FIGURE 2. Module and circuit configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 202640-19T13:35Z
Check the source to verify that this is the current version before use.

TABLE I. Module description. 1/ 2/

Circuit configuration	Module designation	
	RFI Unshielded	RFI Shielded
1	A	H
2	B	J
3	C, D	K, L
4	E	M
5	F	N
6	G	P

1/ All dials have black opaque background with white opaque numerals.

2/ Switch modules are not procurable by themselves, but are part of the switch assembly (see table II).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

TABLE II. Switch assembly.

Part No. M22710/29-	Switch station module number									
	1	2	3	4	5	6	7	8	9	10
001	A									
002	A	A								
003	A	A	A							
004	A	A	A	A						
005	A	A	A	A	A					
006	A	A	A	A	A	A				
007	A	A	A	A	A	A	A			
008	A	A	A	A	A	A	A	A		
009	A	A	A	A	A	A	A	A	A	
010	A	A	A	A	A	A	A	A	A	A
011	B									
012	B	B								
013	B	B	B							
014	B	B	B	B						
015	B	B	B	B	B					
016	B	B	B	B	B	B				
017	B	B	B	B	B	B	B			
018	B	B	B	B	B	B	B	B		
019	B	B	B	B	B	B	B	B	B	
020	B	B	B	B	B	B	B	B	B	B
021	C									
022	C	C								
023	C	C	C							
024	C	C	C	C						
025	C	C	C	C	C					
026	C	C	C	C	C	C				
027	C	C	C	C	C	C	C			
028	C	C	C	C	C	C	C	C		
029	C	C	C	C	C	C	C	C	C	
030	C	C	C	C	C	C	C	C	C	C
031	D									
032	D	D								
033	D	D	D							
034	D	D	D	D						
035	D	D	D	D	D					
036	D	D	D	D	D	D				
037	D	D	D	D	D	D	D			
038	D	D	D	D	D	D	D	D		
039	D	D	D	D	D	D	D	D	D	
040	D	D	D	D	D	D	D	D	D	D
041	E									
042	E	E								
043	E	E	E							
044	E	E	E	E						
045	E	E	E	E	E					
046	E	E	E	E	E	E				
047	E	E	E	E	E	E	E			
048	E	E	E	E	E	E	E	E		
049	E	E	E	E	E	E	E	E	E	
050	E	E	E	E	E	E	E	E	E	E

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:35Z
Check the source to verify that this is the current version before use.

TABLE II. Switch assembly - Continued.

Part No. M22710/29-	Switch station module number									
	1	2	3	4	5	6	7	8	9	10
051	F									
052	F	F								
053	F	F	F							
054	F	F	F	F						
055	F	F	F	F	F					
056	F	F	F	F	F	F				
057	F	F	F	F	F	F	F			
058	F	F	F	F	F	F	F	F		
059	F	F	F	F	F	F	F	F	F	
060	F	F	F	F	F	F	F	F	F	F
061	G									
062	G	G								
063	G	G	G							
064	G	G	G	G						
065	G	G	G	G	G					
066	G	G	G	G	G	G				
067	G	G	G	G	G	G	G			
068	G	G	G	G	G	G	G	G		
069	G	G	G	G	G	G	G	G	G	
070	G	G	G	G	G	G	G	G	G	G
071	H									
072	H	H								
073	H	H	H							
074	H	H	H	H						
075	H	H	H	H	H					
076	H	H	H	H	H	H				
077	H	H	H	H	H	H	H			
078	H	H	H	H	H	H	H	H		
079	H	H	H	H	H	H	H	H	H	
080	H	H	H	H	H	H	H	H	H	H
081	J									
082	J	J								
083	J	J	J							
084	J	J	J	J						
085	J	J	J	J	J					
086	J	J	J	J	J	J				
087	J	J	J	J	J	J	J			
088	J	J	J	J	J	J	J	J		
089	J	J	J	J	J	J	J	J	J	
090	J	J	J	J	J	J	J	J	J	J
091	K									
092	K	K								
093	K	K	K							
094	K	K	K	K						
095	K	K	K	K	K					
096	K	K	K	K	K	K				
097	K	K	K	K	K	K	K			
098	K	K	K	K	K	K	K	K		
099	K	K	K	K	K	K	K	K	K	
100	K	K	K	K	K	K	K	K	K	K

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

TABLE II. Switch assembly - Continued.

Part No. M22710/29-	Switch station module number									
	1	2	3	4	5	6	7	8	9	10
101	L									
102	L	L								
103	L	L	L							
104	L	L	L	L						
105	L	L	L	L	L					
106	L	L	L	L	L	L				
107	L	L	L	L	L	L	L			
108	L	L	L	L	L	L	L	L		
109	L	L	L	L	L	L	L	L	L	
110	L	L	L	L	L	L	L	L	L	L
111	M									
112	M	M								
113	M	M	M							
114	M	M	M	M						
115	M	M	M	M	M					
116	M	M	M	M	M	M				
117	M	M	M	M	M	M	M			
118	M	M	M	M	M	M	M	M		
119	M	M	M	M	M	M	M	M	M	
120	M	M	M	M	M	M	M	M	M	M
121	N									
122	N	N								
123	N	N	N							
124	N	N	N	N						
125	N	N	N	N	N					
126	N	N	N	N	N	N				
127	N	N	N	N	N	N	N			
128	N	N	N	N	N	N	N	N		
129	N	N	N	N	N	N	N	N	N	
130	N	N	N	N	N	N	N	N	N	N
131	P									
132	P	P								
133	P	P	P							
134	P	P	P	P						
135	P	P	P	P	P					
136	P	P	P	P	P	P				
137	P	P	P	P	P	P	P			
138	P	P	P	P	P	P	P	P		
139	P	P	P	P	P	P	P	P	P	
140	P	P	P	P	P	P	P	P	P	P

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Construction: D.

Indexing: 22 or 36 degrees. See truth tables in figure 2.

Temperature-life characteristic: B.

Vibration grade: 3 (15g. 10-2,000 Hz).

Shock type: M (100g), Method I of MIL-S-22710 is applicable.

Altitude rating: 0 (under 10,000 feet).

Terminal design: See figure 2.

Acceleration: Applicable.

Stop Strength: 5 pounds minimum.

Moisture resistance: Method II of MIL-S-22710.

Operating force (tangential): 2 ounces minimum, 10 ounces maximum.

Explosion: Applicable.

EMI/RFI Shielding: Applicable, where specified.

Life (rotational): Method 206, test condition G, MIL-STD-202 (see table III).

TABLE III. Electrical test load.

Test or condition	Resistive	
At atmospheric pressure	.050 Ampere	28 V dc
Life (low level)	10 mA	30 mV dc

Military part number: See tables I and II.

Material (body and dial): Thermoplastic per MIL-P-24519, type GPT-20F or equivalent.

Qualification: For qualification, circuit configuration G of figure 2 shall be used for the test sample.

CAUTION: Isopropyl alcohol recommended for cleaning and solder-flux removal. Avoid the use of acids, ammonia, chlorinated or aromatic hydrocarbons, and oxygenated solvents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, SM
Navy - AS
Air Force - 17
DLA - ES

User activities:
Army - ME
Navy - MC, OS
Air Force - 11

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5930-1176-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-S-22710/29

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:35Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
ELEX 8111
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20363

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:35Z
Check the source to verify that this is the current version before use.